

UV-assisted 3D printing of polymer composites from thermally and mechanically recycled carbon fibers

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Abstract: The global interest in carbon fiber reinforced polymers is going to increase in the next years. Due to the low cost-effectiveness of carbon fibers, the applications are limited to high-performance sectors such as aerospace and automotive. For this purpose, the recovery of carbon fibers has become a well-established practice that also reduces the environmental impact of virgin fibers. Although the 3D printing of thermoplastic carbon fiber reinforced polymers is constantly spreading, the use of recycled carbon fibers in 3D printing is still an unexplored field, especially for composites with a thermosetting matrix. This paper aims to demonstrate the feasibility of recycled carbon fibers 3D printing via UV-assisted direct ink writing. Pyrolyzed recycled carbon fibers with a sizing treatment were firstly shredded to be used as a reinforcement for a thermally and photo-curable resin. Thanks to the addition of a rheology modifier and the customization of a commercial 3D printing apparatus, a batch of tensile specimens were successfully 3D printed. These results will therefore pave the way for the use of recycled carbon fiber reinforced polymers in new fields of application.

Keywords: Polymer-matrix composites (PMCs); crosslinking; resins; additive manufacturing; recycling; carbon fibers; rheology; mechanical testing.

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1. Introduction

Over the last decade, the use of 3D printing to manufacture thermoplastic composites has made remarkable progress in terms of processable materials, printability, and types of loaded fillers [1,2]. To date, a variety of polymer composites have been 3D printed to meet the requirements of a huge range of applications such as healthcare [3], aerospace and automotive sectors [4,5], textile and fashion related fields [6], and packaging [7]. Among these, polymer composites loaded with carbon fibers (CFs) stand out for their good mechanical properties, and they have been recently 3D printed using a thermoplastic matrix by the technology of Large Format Additive Manufacturing [8]. When compared to other conventional technologies, the additive manufacturing of fiber-reinforced polymers offers the added advantage of providing desired complex shapes and geometries to obtain lightweight and robust objects without heavily increasing the production costs [9].

In addition to 3D printed thermoplastic polymers loaded with CFs, UV-assisted three-dimensional (UV-3D) printing technology has been recently used to build 3D scale models of complex structural components such as airfoils and propellers [5]. Using the

same approach, 3D freeform and freestanding nanocomposites were successfully fabricated to produce cellular nanocomposite materials with hexagonal or squared unit cells [10] and wearable textile, smart and strain micro-structured sensors [11,12]. However, the high cost of CFs often limits their use in some advanced and high-performance applications, such as aerospace and automotive fields. The aerospace industry is the most important sector for CFRP with 32% of the total global demand, followed by the automotive industry, which currently represents around 21.8% of the total demand. Despite the limitations of CFs, the global demand for carbon fiber reinforced polymers (CFRP) was estimated at 141,000 tons in 2019, with an average annual growth rate of about 10%, leading to a forecast of around 170,000 tons in 2021 [13].

To improve the cost-effectiveness of CF reinforced polymers (CFRPs) and to reduce their environmental impact, the development of CFRPs recycling technologies is of crucial importance [14,15]. The recovery of carbon fibers from composites is a well-established practice and the pyrolysis of CFs has been classified so far as the most mature technology for the recycling of CF composites [14]. Moreover, the recovery of CFs through pyrolysis is estimated as having a much lower energy demand and carbon footprint than virgin CFs production (i.e. more than 80% decrease in the cumulative energy demand and more than 50% reduction of greenhouse gas emissions) [16]. However, recycling processes can reduce the surface and tensile properties of recycled CFs (rCFs) [17]. It is therefore important to explore new routes of reuse for rCFs and investigate the resulting performance of reprocessed CF composites.

Here, in this work, pyrolyzed carbon fibers underwent a sizing treatment, designed to reuse them for the realization of recycled thermoplastic composites. For a circular economy model, these already-sized CFs can offer the opportunity to reuse recycled CFRPs in different applications and extend their lifetimes. In this perspective, the pyrolyzed and sized recycle was shredded and combined with a thermally and UV-curable resin to demonstrate the processability of rCF thermosetting composites through UV-assisted direct ink writing (DIW) process. To the best of our knowledge, this is a pioneering work, which describes the use of rCFs in the 3D printing process. This work is of great importance as it demonstrates the possibility of 3D printing self-standing structures loaded with rCFs. Although the 3D printable resin loaded with rCFs showed a low degree of UV-induced crosslinking during the DIW process, self-standing structures were successfully 3D printed thanks to the addition of a rheological modifier. A urea-modified agent was selected as a rheological modifier, taking into account previous works, where self-standing objects were 3D printed due to the formation of a hydrogen bond structure [18]. The present work shows a successful 3D printing of a curable resin loaded with recycled CFRPs. The selection of an appropriate modifier of the 3D printable resin and the use of a UV-assisted DIW process enabled the additive manufacturing of CF composites with a high elastic modulus in agreement with the theoretically expected value. A fully cross-linked material was also obtained with a post-curing cycle after printing. This study paves the way for future research activities to explore the printability of freeform and complex structures as well as improve fiber-matrix adhesion and thus mechanical properties.

2. Materials and Methods

2.1. Materials

The composite matrix consisted of ethoxylate bisphenol A diacrylate resin supplied from Arkema, Colombes, France (local distributor: Came S.r.l., Lainate, Italy), and named SR349.

To initiate the polymerization by UV light, ethyl phenyl (2, 4, 6-trimethyl benzoyl) phosphinate, hereinafter named TPO-L, was added to the acrylic resin (Lambson Limited, Wetherby, United Kingdom). Dicumyl peroxide was also added to the matrix to increase the crosslinking degree during the thermal post-curing (Sigma-Aldrich Corporation, St.

Louis, Missouri, US). The crosslinking agents were added to the solution with a proportion of 3 wt% and 0.3 wt% respectively.

A solution of a urea-modified agent in amide ester (BYK-7411 ES from Sigma-Aldrich Corporation, and hereinafter called BYK) was added to the resin system as a rheological modifier to improve the quality of the extrusion during the process. Raw components for the resin system were used as received.

2.2. Pyrolysis and Sizing

Recycled CFs were obtained from a pyrolysis process performed by Tecnia Research & Innovation Center (San Sebastian, Spain) on expired pre-impregnated (prepreg) CFs, kindly supplied by Aernnova Aerospace S.A (Minano Mayor, Spain).

For the rCF resizing, a non-ionic maleated polypropylene dispersion (Hydrosize® PP2-01, Michelman Inc.) of 5 wt% solids content with deionized water was generated. The dispersion was stirred at 30 °C and for 1 h for good dispersion. The rCFs were immersed into the solution and then dried in a convection oven at 80 °C for 10–12 h and sealed in plastic bags to wait for the shredding. This procedure without any rinsing of excess sizing generates fiber agglomerates, which have a higher density than the untreated rCFs and are thus more suitable for shredding. However, the large amount of sizing on the CF surface might absorb moisture before compounding and act as a plasticizer at the fiber-matrix interface.

2.3. Shredding

To 3D print, the thermally recycled and sized CFs, their length had to be reduced. Firstly, a shredding step with a cutting mill by Retsch GmbH (model SM-300, Haan, Germany) at a velocity of 2000 rpm with a grate size of 1 mm was performed. Figure 1a and b show the breaking chamber of a cutting mill (Retsch model SM300) and an example of a cutting line composed of six different tungsten carbide inserts. However, after this shredding step, it was not possible to obtain a powder material to be used for 3D printing, as shown in Figure 1c.

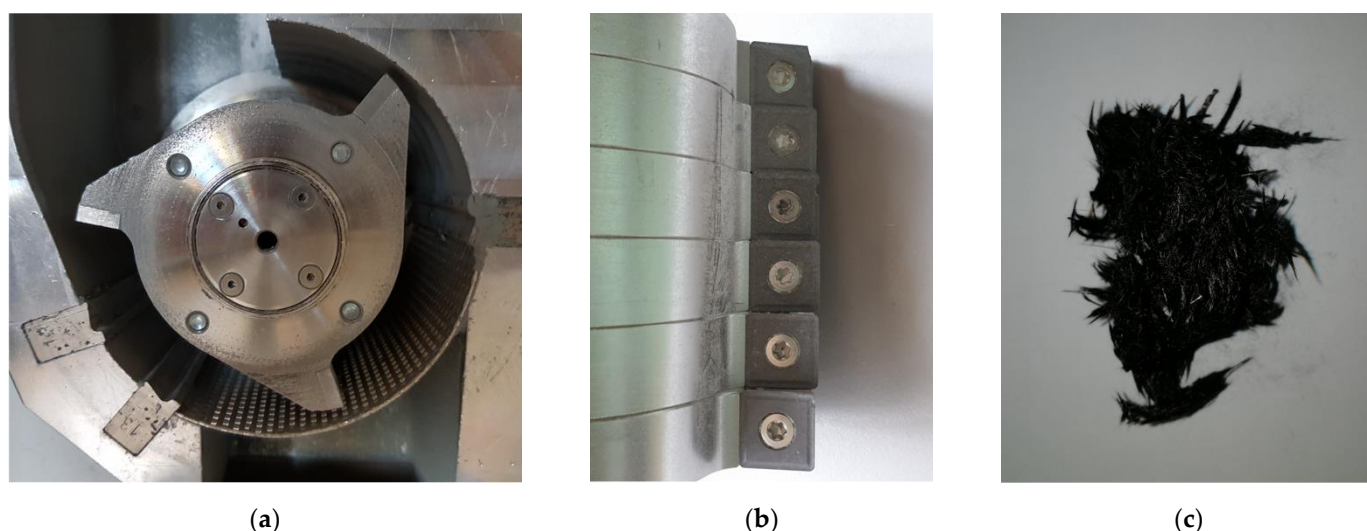


Figure 1. (a) Shredding chamber of a Retsch SM300 cutting mill; (b) a particular of a line composed by six tungsten carbide inserts in the middle and an image of thermally rCFs after the thermal recycling; (c) sizing and first shredding with a cutting mill on the right.

A further shredding step was performed to reach the dimensional requirements for the UV-DIW process [19]. To further reduce the rCF length, a quad blade chopper CH580 (Kenwood Limited, Havant, United Kingdom) was used. The first attempt was carried with 1 g of material alternating 1 min of chopping and 30 s of pause for a total of 30 min.

After this first chopping, no significant difference was visible when compared to Figure 1c. To enhance the brittleness of rCFs, dry ice was inserted in the chopper, and the process was repeated with a longer cutting period, resulting in 5 min chopping and 1 min pause for a total period of 2 h. To separate the different fiber sizes obtained, a manual sieving process with 300 and 100 μm sieve mesh size was performed. Figure 2a shows the fraction with a dimension higher than 300 μm because it could not be sieved. Figure 2b and c show the two particle sizes obtained with the sieving process.

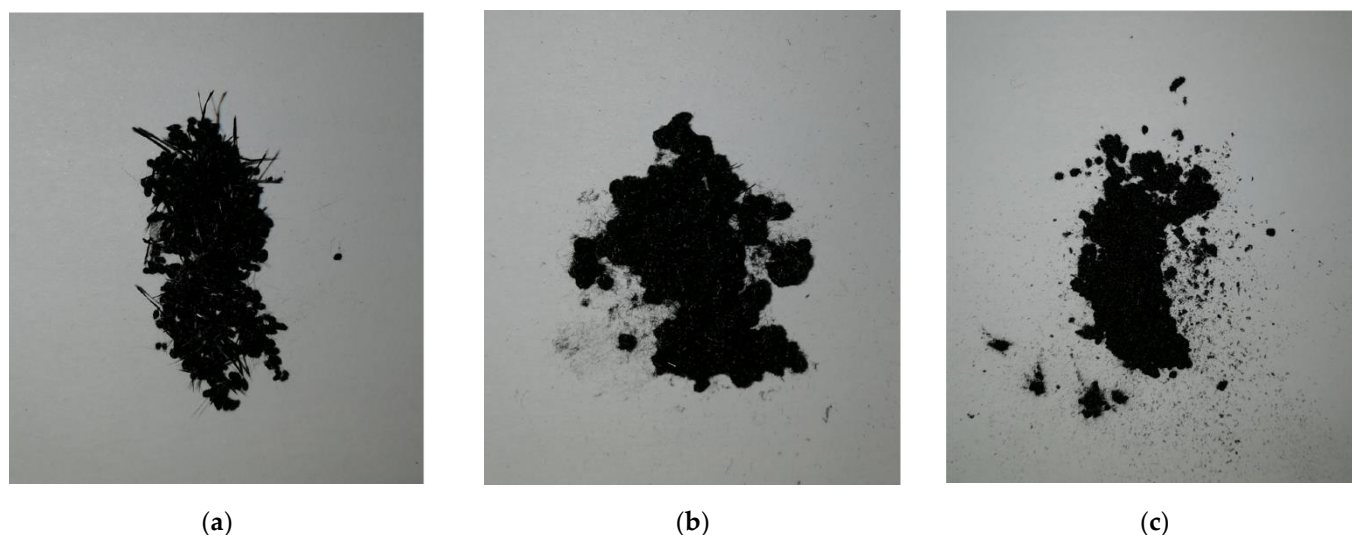


Figure 2. rCFs images after the shredding process and manual sieving: (a) rCFs fraction with a dimension higher than 300 μm ; (b) rCF fraction with a dimension higher than 100 μm ; (c) rCFs fraction with a nominal dimension of 100 μm .

2.4. 3D Printing

3D printable ink formulations were obtained by mixing the resin system with a magnetic stirrer at room temperature for 2 h. Afterward, rCFs were manually incorporated into the matrix to obtain a homogeneous and printable material.

For the sake of simplicity, a nomenclature was assigned for each 3D printable ink formulation according to the “XBYAER” scheme. Specifically, X corresponds to the urea-modified agent percentage in weight, and it is followed by the first letter of its commercial name “BYK”. Y represents the filler concentration by weight of the rCFs, and it is followed by the short name “AER” of the primary supplier Aernnova. Accordingly, 6B25AER formulation contains 6 wt% of the urea-modified agent and 25 wt% of rCFs.

After the mixing, 3D printable inks were suitable for the DIW process. The starting 3D printing apparatus was based on a modified version of a commercial FDM 3D printer (3Drag supplied by Futura Group S.r.l., Gallarate, Italy) and it was described in a previous work [20]. Considering the photo-polymerization issues related to CF, other modifications were made to the system, and they will be better described in the next paragraph. For this purpose, Fusion 360 (Autodesk, San Rafael, California, US) was the main CAD software adopted for the design of the custom pieces. Prusa i3 MK3S FDM 3D printer was then used for 3D printing the models by creating a Gcode file with Slic3r PE 1.41.3 open source slicing software (Prusa Research, Prague, Czech Republic). For tensile specimens performed with DIW, Cura open source slicing software was adopted (Ultimaker B. V., Utrecht, The Netherlands).

For the present experimentation, a 20 mL syringe with a stainless-steel conic UV-shielded nozzle (1.04 mm diameter) was equipped for the system. The parameters employed during the 3D printing are listed in Table 1.

Table 1. 3D printing parameters for rCF composites with 20 wt% of rCF and sided LEDs position.

Parameters	6B20AER
Perimeter number (-)	0
Infill (%)	100
Raster angle (°)	0
Flow (%)	100
Speed (mm/s)	12 – 15
Layer Height (mm)	0.20
Nozzle Diameter (mm)	1
LED UV (-)	3 x 3 W (395 nm - lateral)

The polymerization conversion of the material should be increased using a post-curing cycle, taking into account the CF light absorption. For this reason, a UV post-curing step was carried out in a UV chamber Polymer 500 W (Helios Italquartz S.r.l., Cambiago, Italy) with a UVA emittance mercury vapor lamp type Zs (950 W/m²) for 15 min for each sample side. Afterward, a thermal post-curing step was performed in a non-controlled atmosphere oven at 140°C for 2 h.

2.5. Material Characterization

The recycle morphology was deeply investigated using Scanning Electron Microscopy (SEM) micrographs performed with Cambridge Stereoscan 360 (Cambridge Instrument Company Ltd., Cambridge, United Kingdom). Secondary and backscattered electron probes were used, and physical vapor deposition of gold for 1 min was performed to prepare the sample surfaces.

Furthermore, SEM micrographs were also employed for the evaluation of the carbon fiber length and diameter through MATLAB software (The MathWorks Inc., Natick, Massachusetts, US). The resulting data were used to calculate the aspect ratio distribution. SEM images of AER recycle and data from the MATLAB evaluation are visible in the Appendix. The aspect ratios of the fibers were also used for the calculation of the Halpin-Tsai modulus prediction.

Rheological tests on rCFs were carried out with Discovery HR-2 (TA Instruments Inc, New Castle, Delaware, US) with a 20 mm plate-plate geometry and a 1 mm gap. A 3-step oscillation strain test was performed at 1 Hz (forerun by only 3 minutes rest period), 0.01% strain for 20 s, 20% strain for 20 s, and 0.01% strain for 1 h.

Additional analyses were performed for a better understanding of the degree of cross-linking of the 3D printable ink. The glass transition temperature of the 3D printed sample after post-curing was evaluated through differential scanning calorimetry (DSC) with a Mettler-Toledo DSC/823e (Mettler Toledo, Columbus, Ohio, US). The heating ramp was set from 0°C to 250°C with a 20°/min of heating rate.

Gel content was also evaluated to determine the quantity of cross-linked material after the post-curing treatment. After the immersion of the material in acetone for 24 h on a magnetic stirrer (with a proportion of 300 ml for 2 g), the solution was filtered, and not solubilized residual particles were collected in a paper filter (Whatman Filter Papers, Cat No 1001 125). The filtered solution was poured into a flask. The solid residue and the flask were put in a vacuum oven at 50°C for approximately 48 h, until two consecutive weightings, after 6 h from each other, were constant. Knowing the weight of the flask and the paper filter, the final weights of the solubilized residue in the flask and the not solubilized one can be calculated, respectively. Finally, Gel percentage (Gel%) can be estimated thanks to the equation:

$$Gel\% = \frac{m_{sample} - m_{sol}}{m_{sample}} \times 10 \quad (1)$$

where m_{sample} is the starting mass of the material sample, and m_{sol} is the mass of the solubilized residue after the procedure.

Tensile properties were evaluated with a Zwick Roell Z010 (ZwickRoell GmbH & Co. KG, Ulm, Germany) with a 10 kN cell load at a speed of 1 mm/min. For the tests, the ASTM standard test method D3039/D3039M-17 (2017) was adopted [21]. Accordingly, specimens were 3D printed with a nominal gauge length of 40 mm, a width of 10 mm, an overall length of 100 mm, and a thickness of 2 mm. Then, they were manually polished to remove the asperities and achieve a constant cross-sectional area, and their final dimension was measured. Afterward, experimental elastic moduli values were compared with the Halpin-Tsai model for aligned fibers [22]. The prediction was performed by calculating the theoretical elastic modulus for each measured fiber. Afterward, the values were averaged to obtain the Halpin-Tsai elastic modulus and its standard deviation.

3. Results and Discussion

3.1. Shredding

The objectives of the shredding process (also called size reduction, grinding, or comminution) are mainly two. The first one is to obtain high liberation of target materials from heterogeneous particles. Higher degrees of liberation are achieved by a particle made of a small number of materials and the highest possible liberation is obtained for particles made of only one material [23]. This is particularly important in particles with high-value materials that will be treated with the following separation steps. The second objective is to create homogenous particle flows (both in shape and in size) with fine controlled dimensions. This is fundamental for the direct reuse of these particles in new high added-value products as those presented in this paper.

Depending in particular on the initial dimensions of end-of-life products, it could be possible to have several size reduction steps that could be divided into coarse and fine shredding. Coarse ones reduce dimensions of products to 10–20 mm, obtaining particles that, in most cases, are not liberated. On the other hand, fine comminution can take in input coarse particles giving in output particles with a dimension from 6 mm to few tens of microns (in case of ultrafine size reduction processes).

To reduce particle dimensions, it is possible to apply mechanical or non-mechanical strains (either thermal or chemical ones). Due to the low cost and reduced energy consumption, this work focuses on mechanical ones, based on pressure, shear, cut, or impact (both interparticle or with a surface) or a combination of them. Table 2 compares the different technologies commonly used for shredding of composite materials with an indication of the throughput, underling the efficiency and highlighting the suitability for composites.

Table 2. Comparison of the different shredding technologies commonly used for composites comminution.

Technology	Type of Shredding	Throughput	Efficiency	Suitability for Composites
Two shafts shear shredder	Coarse	High	Risk of fibers wrapping	Good for coarse shredding
Single shaft shear shredder	Coarse	High	Good	Good for coarse shredding
Hammermill	Coarse/fine	High	Good for thermoset matrixes, not suitable for thermoplastic matrixes	Good for coarse shredding of thermoset composites
Impact crusher	Coarse	High	Not suitable for thermoplastic matrixes	Low
Jaw crusher	Coarse	High	Not suitable for thermoplastic matrixes	Low
Cutting mill	Fine	Mid	Good	Good for fine shredding

Disc mill	Fine	Low	Not suitable for thermoplastic matrixes	Low
Chain mill	Coarse	High	Good for thermoset matrixes, not suitable for thermoplastic matrixes, creation of powder	Good for coarse shredding of thermoset composites

Among all the different technologies in the table, the most suitable one for shredding of composite materials is the cutting mill, in particular considering continuous feed mode, enabling profitable industrial applications. Firstly, this machine allows the treatment of both thermosets and thermoplastics composites, in particular, if paired with a cyclone system (facilitating the exit of the particles from the shredding process and decreasing the internal temperature of the comminution chamber). In addition, the cutting mill, in continuous feed mode, can lead to a homogeneous distribution of particles in terms of both shape and dimension. This is possible thanks to the combination of cut and compression strains and particles discharge through a grate with a specific controllable dimension. A fast-rotating rotor (from 700 rpm to 3000 rpm) is equipped with multiple removable inserts, usually organized in different lines. The material is loaded from the top of the chamber and is simultaneously cut by the inserts and compressed on the chamber. The material is unloaded from the bottom through a grate of variable size. Cutting, shearing, and impact effects cooperate for a homogeneous and effective size reduction of materials. This configuration is ideal for heterogeneous materials, which can be hard cut to obtain a fine size (down to 0.25 mm). This process requires typically high forces to break particles, leading to an increase in temperature in the chamber. As a consequence, highly malleable, soft, and rubbery materials may deform, slowing down the comminution process or, in worst cases, blocking the machine. Nevertheless, the availability of hard fibers in composites mitigates the risk of slowing down due to the lamination effect.

Several factors affect process results with the cutting mill, both design and controllable parameters. The first ones are the volume of the chamber, influencing the maximum quantity of material that could be treated and the consequent throughput, the number of cutting tools, and the number of cutters per cutting tool, affecting the efficiency of the breakage process. On the other hand, the controllable parameters are the rotational speed of the rotor, impacting on the throughput and the cost of the process, the feed rate, strictly related to the throughput and the saturation of the chamber, and the grate size. This last one is the most important to reach a specific target dimension, maximizing the number of particles around a specific size.

However, after the first shredding process was performed with the cutting mill, the thermally recycled carbon fibers showed higher dimensions than the nozzle used for 3D printing. To meet the dimensional requirements for the 3D printing of reinforced composites, preliminary tests were performed with a quad blade chopper to obtain a finer recycled material. This result was already shown in Figure 2b and c. The obtained rCFs were successively measured through SEM micrographs. Micrographs of the rCFs can be seen in the Appendix together with the measurement results in terms of fiber length, diameter, and aspect ratio distribution (Figure A1 and Table A1). As described in the Experimental Section, the aspect ratio distribution was also used for the Halpin-Tsai model calculations.

3.2. Ink formulation and process requirements

As described in our previous work [24], the material can be crosslinked during the 3D printing process thanks to three UV-LEDs controlled by an external unit. Consequently, the presence of a UV photo-initiator is needed. To allow the layer-by-layer deposition, the UV crosslinking of the material has to be rapid enough to ensure the shape retention of the 3D printed material.

Since the resin matrix is a dual-cure system, the final degree of crosslinking will be the sum of the UV and thermal crosslinking. The UV source can be employed only to stabilize the deposited material during the fabrication of the 3D printed objects, whereas the thermal post-curing aims to finalize the material crosslinking. As shown in Figure 3, the UV conversion of the neat resin system and the ink containing the rCF before and after the 3D printing process was evaluated by UV-DSC.

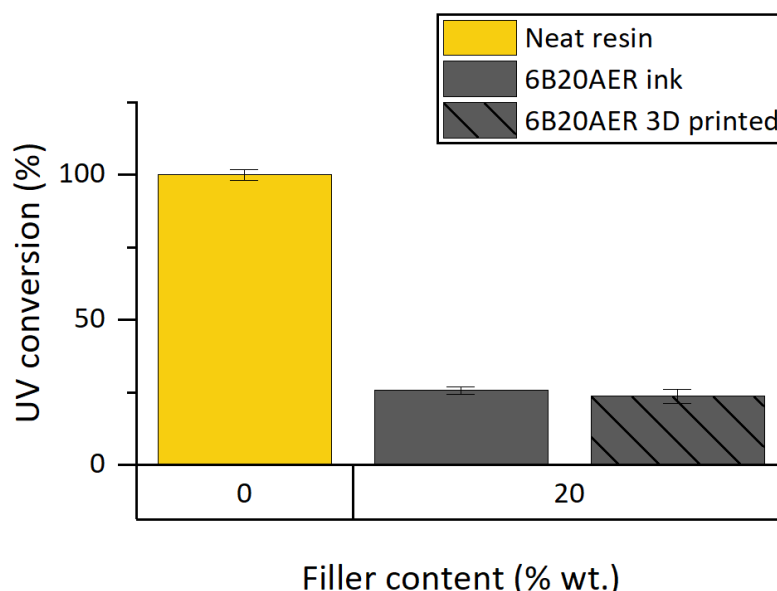


Figure 3. Characterization of the UV-induced crosslinking of rCF 3D printable inks: UV conversion of 3D printable inks and 3D printed specimens evaluated by UV-DSC.

Considering the UV conversion of the neat resin as the maximum, the ink has a very low conversion. Moreover, the conversion during the fabrication process is almost negligible since there is very little difference between the UV-DSC analysis for the ink before and after the 3D printing process. For these reasons, a change of the rCF ink formulation and a modification of the process were necessary.

3.3. Rheological modification

To overcome the poor UV reactivity of the rCF-based 3D printable ink, a rheological modification was studied with the intent to achieve a correct three-dimensional deposition even with a low conversion material. This allowed shape retention to be achieved not only by the UV conversion during the 3D printing process but also by the rheological modifier into the ink formulation.

The addition of a modified-urea agent was a solution already seen in the literature [18]. The influence of this rheological modifier was tested for rCF 3D printable ink, and the results are shown in Figure 4. The test was specifically defined to be comparable with the 3D printing process. In detail, the first step of the test shows the material inside the reservoir (oscillation strain quasi-zero), the second step of the test shows the material during extrusion (the oscillation strain was fixed at 20%). Finally, the third step shows the material behavior after the deposition.

The addition of 6 wt% over the resin system allows for the transition between solid-like and liquid-like behavior of the ink during extrusion. Figure 4a and c show, respectively, the difference of the ink without and with the modified-urea agent, highlighting how its addition modified the ink behavior.

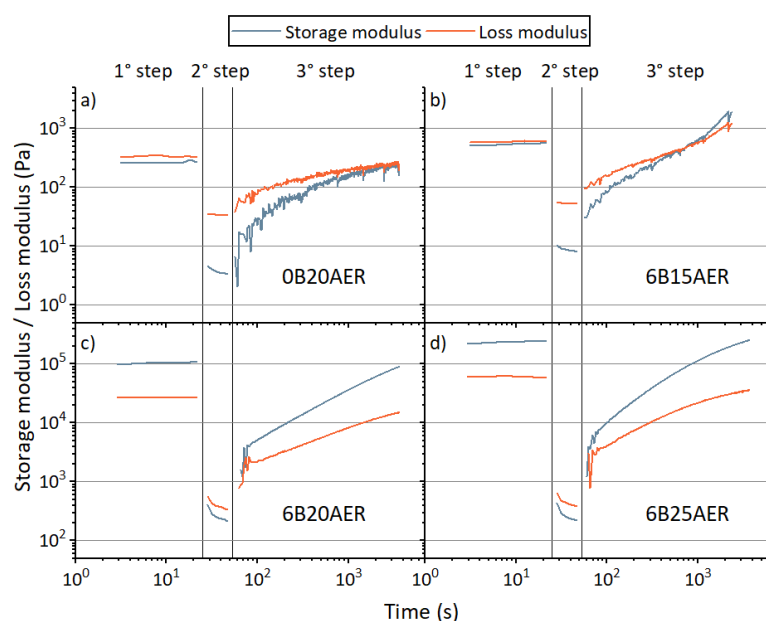


Figure 4. Rheological step tests for CF fine recycle 3D printable inks.

At rest, before and after extrusion, the ink behaves like a solid. As a consequence, the deposited material is stable and it will not lose the shape defined through the deposition process. Figure 4b, c, and d show the influence of different percentages of rCFs over the rheological behavior. Reducing the rCF content to 15 wt% (6B15AER), the solid-like behavior of the ink is not rapidly recovered after the extrusion, as shown in Figure 4b where G'' is higher than G' at the beginning of the third step of the test. The material starts to recover the solid-like behavior only after 15 min (≈ 1000 s) of rest. Moreover, 6B15AER ink behaves like a fluid at the first step of the test. This shows that the stresses and deformations that occurred during the material loading in the rheometer plates destroyed H-bonding interactions in 6B15AER ink with the urea-based rheological modifier. The time passed before starting the experiment was not enough to reform hydrogen bonds. Increasing the rCFs content, the change from the liquid-like to solid-like behavior occurs rapidly, and the transition timeframe is comparable with the deposition process.

Considering the poor UV conversion of the material and the low difference in solid-like recovery after extrusion between 20 wt% and 25 wt% rCFs inks, 20 wt% rCFs ink was selected (6B20AER ink).

3.4. Crosslinking and thermal properties

The material was tested to assess the post-curing cycle performance. DSC analysis and gel content were evaluated. As shown in Table 3, the calorimetry analysis shows a residual enthalpy of 11.4 J/g after the post-curing cycle and a glass transition temperature of 82 °C.

Table 3. DSC and Gel% results for rCF samples

3D printable ink	Residual Enthalpy (J/g)	T _g (°C)	Gel% (%)
6B20AER	11.4	82	99

The presence of a residual enthalpy could be due to unreacted crosslinking agents. Alternatively, it could be linked to the reduced mobility of the partially crosslinked material that hinders the propagation of the reaction. Nevertheless, the gel content measure-

ment evidenced a high degree of crosslinking, confirming the good quality of the cross-linked material. Further investigations could give a more accurate explanation of the presence of possible unreacted groups.

3.5. 3D Printing process

As anticipated in the process requirements section, a modification of the process was investigated. The previous 3D printer setup allows the deposition of the material and, contextually, its in-situ polymerization. This solution performed well with high reactivity inks, but it led to a difficult control over the 3D printed shape for inks with a low UV conversion. Figure 5 shows the previous 3D printer apparatus (on the left), and the new one (on the right). The main changes are related to the UV-LED system that was modified and positioned at the side of the extrusion head.

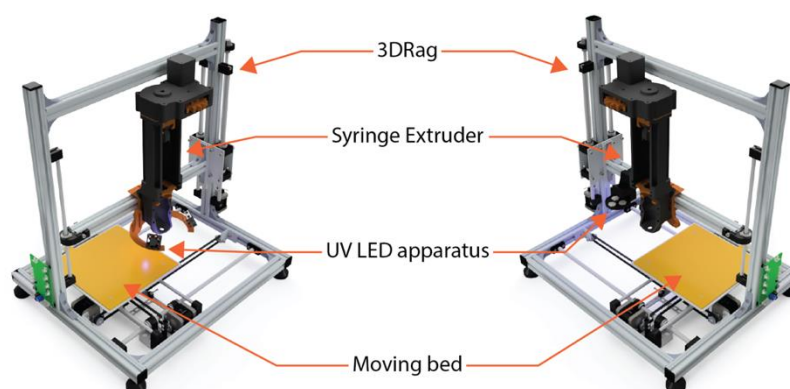


Figure 5. The previous version of the UV-DIW 3DRag printer (on the left, Romani et al.[20]) and the new 3D printing layout for CF reprocessing (on the right).

After the deposition of each layer, the curing apparatus was moved over the deposited material and switched on. To crosslink the material conformingly, the UV-LED apparatus was forced to move over the 3D printed object following the deposition path. This was obtained by manual modification of the Gcode file that consisted of translating the extrusion paths by a distance equal to the horizontal offset between the center of the UV-LED apparatus and the tip of the extrusion head. Figure 6a and b show the deposition step and the curing step respectively. The generation of the Gcode file can be optimized for complex objects which require longer Gcode files. However, for the generation of the Gcode in this work, a manual modification of the file was performed because it was easier due to the simple deposition path needed for the fabrication of a tensile specimen. The result of the 3D printing process is shown in Figure 6c. Five tensile test specimens were successfully fabricated.

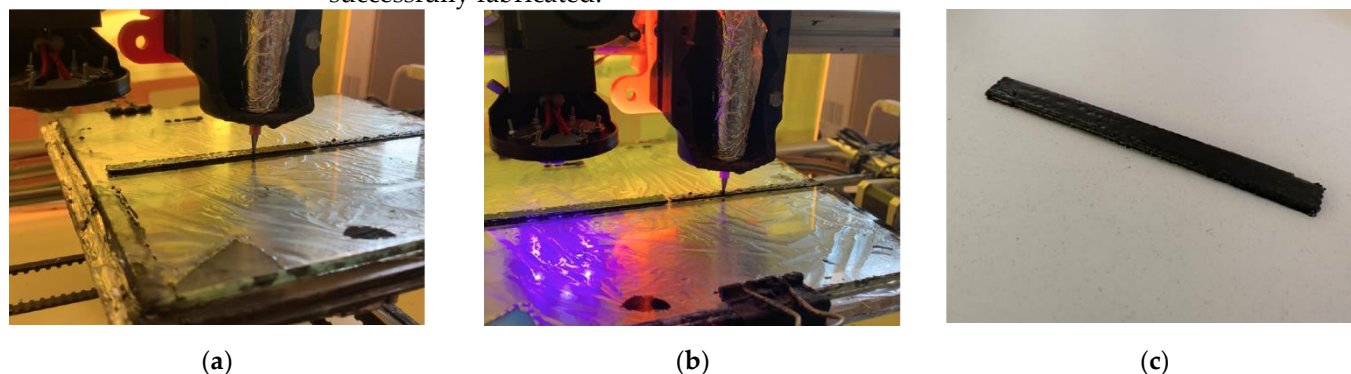


Figure 6. (a) 3D printing phase with the modified side-LED apparatus; (b) UV conversion phase; (c) 3D printed tensile specimen with 20 wt% of rCFs.

3.6. Mechanical properties

Finally, the tensile properties were tested. The averaged stress vs strain curve measured for the rCFs remanufactured material is shown in Figure 7 together with the averaged curve of the neat resin.

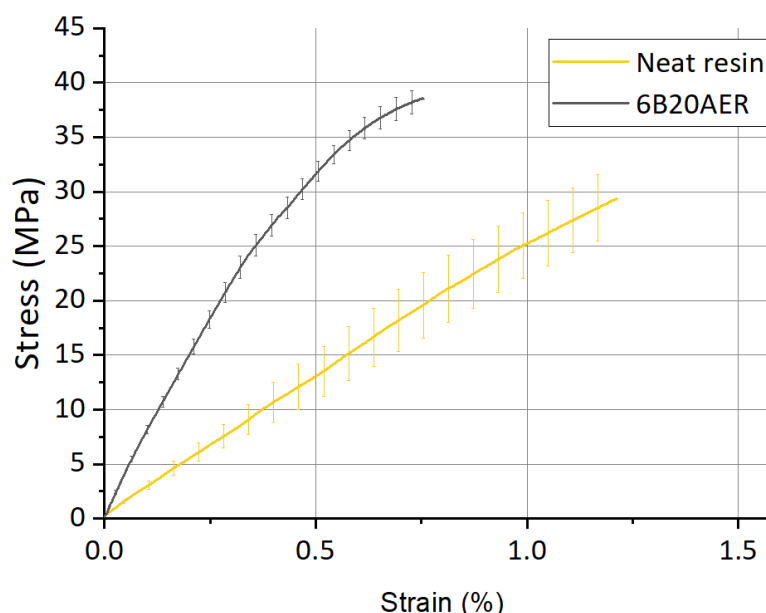


Figure 7. Mechanical behavior of 20 wt% CF formulation and neat resin: Stress versus Strain graph (Average curves).

A brittle behavior was observed for both the neat resin and the remanufactured composite. As visible in Figure 7, a quite high increase in stiffness for the rCF composite sample was achieved in comparison with the neat resin. Despite the elongation at break is lower than the neat resin, the tensile strength value increased. The mechanical properties of the two materials are reported in Table 4.

Table 4. Mechanical properties of 20 wt% CF formulation and neat resin: Elastic Modulus and Halpin-Tsai model, Tensile Strength, Elongation at break, and Toughness.

Formulation	Layer Height (mm)	Elastic Modulus (GPa)	Halpin Tsai Production (GPa)	Tensile Strength (MPa)	Elongation at Break (%)	Toughness (J/mm ³)
Neat Resin	//	2.6 ± 0.4	//	25.9 ± 1.2	1.0 ± 0.2	136.6 ± 25.0
6B20AER	0.20	8.1 ± 0.5	8.1 ± 2.2	39.4 ± 2.1	0.9 ± 0.1	218.6 ± 57.2

Due to the relatively low viscosity of the neat resin, tensile test specimens for this material were produced by casting and the results were also reported in a previous work [24]. Relatively thin layers of 0.2 mm were deposited for the rCF composite remanufacturing. The elastic modulus of 3D printed recycled composites was three times higher than that of the neat resin. This result was confirmed also by the Halpin-Tsai theoretical model for aligned fibers, which predicted very well the experimental results. Although the elongation at break was lower than the neat resin, the remanufactured composite material was tougher. The increased rigidity together with the increased tensile strength led to an increase in toughness of 160% for the material.

Fracture surface analysis confirmed the preferential alignment of the fibers, which is longitudinal to the print direction and perpendicular to the fracture surface. The printing direction was in fact longitudinal to the specimen. As shown in Figure 8a, the matrix ho-

mogeneity evidence the absence of 3D printing marks between the deposited layers. However, some voids related to entrapped air bubbles were detected. Figure 8b and c show the exposed fibers with a clean surface and the holes left by the pulled-out fibers. Accordingly, fibers pullout can be stated as the predominant failure mode of the material.

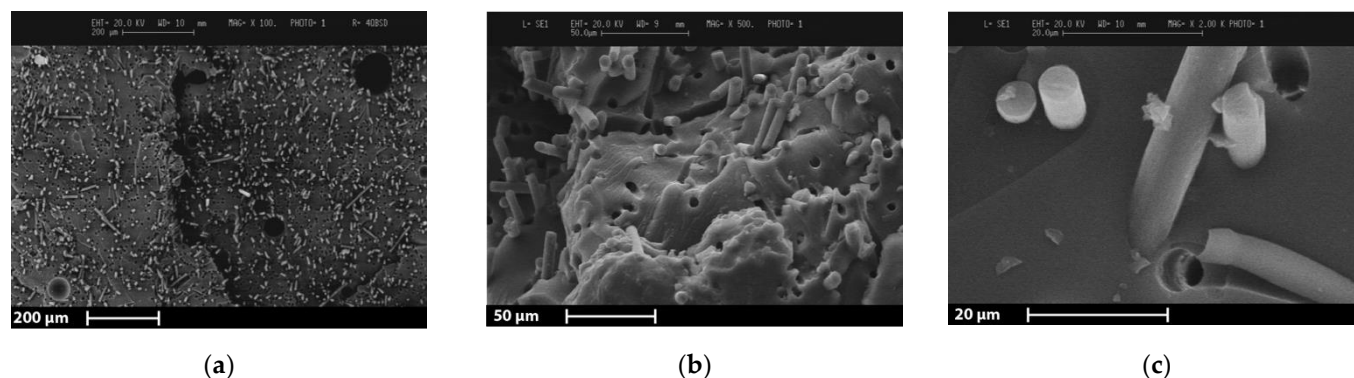


Figure 8. SEM micrographs of 20 wt% CF formulation of 3D printed tensile specimen cross-sections: (a) details on voids and overall fibers distribution in the matrix; (b) fibers detachment from the matrix; (c) fiber interactions with the matrix.

4. Conclusions

This paper demonstrates the feasibility of the 3D printing of rCF thermosetting composites. Thermally recovered CFs can be reprocessed to obtain a reinforcing filler for photo- and thermally curable 3D printable inks. Moreover, this work reveals a high potential for the use of thermally recycled carbon fibers shredded with a frugal approach, even if a systematical design of the shredding process is needed. This will be investigated in future works through an extensive experimental campaign with innovative shredding machines as an ultra-centrifugal comminution system. Thanks to the addition of a urea-modifier agent, the poor UV reactivity of the ink due to the rCF presence is not an obstacle for the UV-assisted 3D printing process. Shape retention was achieved thanks to a new 3D printing apparatus that separates the UV crosslinking phase from the material extrusion for each 3D printed layer. Future works on applications, case study objects, and complex models could further prove the potentiality of decoupling the material deposition by the material crosslinking processes. Considering the mechanical properties, the addition of rCFs into the matrix resulted in a noticeable increase in toughness and elastic modulus. The latter values confirmed the theoretical Halpin Tsai prediction, showing an overall quality of the UV-assisted 3D printing process. Nevertheless, the toughness of the material could be furtherly increased by specifically designing a sizing agent. In this way, the interfacial adhesion of the sized rCFs with the resin used in this work could be higher, leading to better mechanical performance. Although the sizing agent was designed for rCFs composites with a polypropylene matrix, the exploitation of these rCFs was investigated in a circular economy perspective and good mechanical properties were obtained in this paper. From these promising results, 3D printed rCF reinforced thermosetting materials can potentially lead to new technology applications in circular economy models, substantially reducing the economic and environmental impacts of the composite manufacturing sector on a global scale.

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Data Availability Statement: Publicly available datasets were analyzed in this study. The data can be found here:

[https://polimi365-my.sharepoint.com/:f/g/personal/10090710_polimi_it/EqX09fsItn1Eot-QFYbQhrq8BGQFXhGyYZNjn_R9-hzz-IA?e=XBofVU]. If you will use the data, please cite them in the following way: [dataset] Andrea Mantelli, Alessia Romani, Raffaella Suriano, Marco Diani, Marcello Colledani, Essi Sarlin, Stefano Turri and Marinella Levi. 2021. UV-assisted 3D printing of polymer composites from thermally and mechanically recycled carbon fibers; Data repository link.

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Appendix A

SEM micrographs were analyzed with a MATLAB script to measure the length and the diameter of each fiber. The average values of the rCF morphology analysis are reported in Table A1. Figure A1 shows two representative micrographs of the rCFs and their aspect ratio distribution.

Table A1. Average values of carbon fiber length, fiber diameter, and aspect ratio for rCF.

Filler	Length (μm)	Diameter (μm)	Aspect Ratio (-)
rCF	74.6 ± 48.3	6.5	11.5 ± 7.4

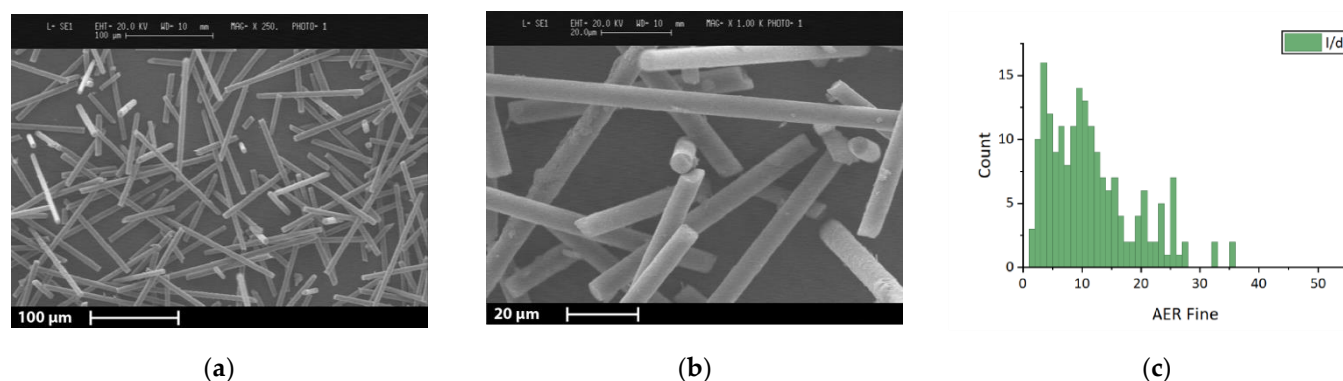


Figure A1. (a) SEM micrographs of rCFs showing the different length; (b) inset of the micrograph; (c) rCF aspect ratio distribution graph.

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